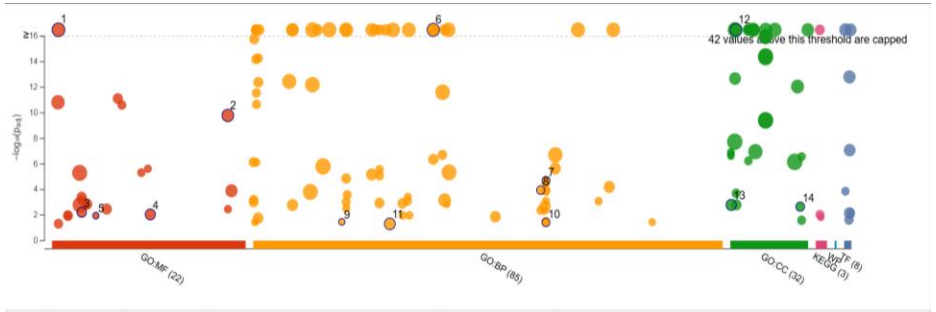
WT and dCIN5 Comparison

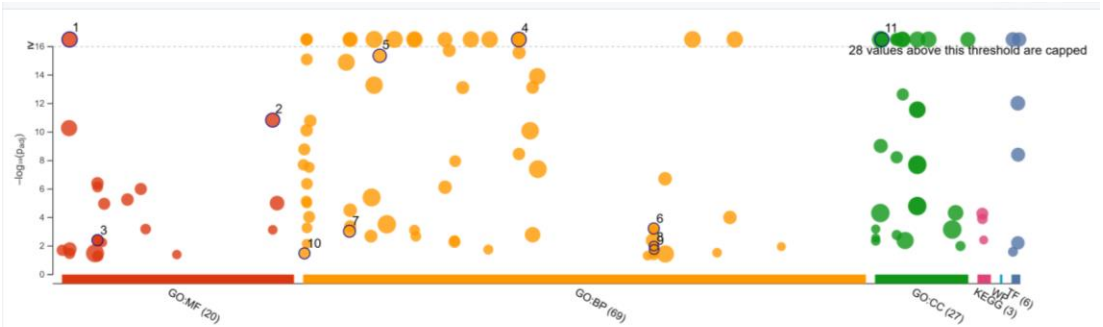
Profile 45 – overview

- WT



ID	Source	Term ID	Term Name	p_{adj} (query_1)
1	GO:MF	GO:0003723	RNA binding	3.888×10^{-22}
2	GO:MF	GO:0140098	catalytic activity, acting on RNA	1.593×10^{-10}
3	GO:MF	GO:0008170	N-methyltransferase activity	6.029×10^{-3}
4	GO:MF	GO:0043021	ribonucleoprotein complex binding	8.919×10^{-3}
5	GO:MF	GO:0015146	pentose transmembrane transporter activity	1.106×10^{-2}
6	GO:BP	GO:0042254	ribosome biogenesis	2.573×10^{-64}
7	GO:BP	GO:0071042	nuclear polyadenylation-dependent mRNA catabol...	1.989×10^{-5}
8	GO:BP	GO:0070478	nuclear-transcribed mRNA catabolic process, 3'-5' ...	1.107×10^{-4}
9	GO:BP	GO:0015750	pentose transmembrane transport	3.397×10^{-2}
10	GO:BP	GO:0071034	CUT catabolic process	1.755×10^{-2}
11	GO:BP	GO:0032543	mitochondrial translation	4.646×10^{-2}
12	GO:CC	GO:0005730	nucleolus	2.759×10^{-72}
13	GO:CC	GO:0000313	organellar ribosome	1.591×10^{-3}
14	GO:CC	GO:1902555	endoribonuclease complex	2.182×10^{-3}

- dCIN5



ID	Source	Term ID	Term Name	p_{adj} (query_1)
1	GO:MF	GO:0003723	RNA binding	3.724×10^{-17}
2	GO:MF	GO:0140098	catalytic activity, acting on RNA	1.443×10^{-11}
3	GO:MF	GO:0008170	N-methyltransferase activity	3.868×10^{-3}
4	GO:BP	GO:0042254	ribosome biogenesis	4.657×10^{-46}
5	GO:BP	GO:0009451	RNA modification	4.430×10^{-16}
6	GO:BP	GO:0071025	RNA surveillance	5.824×10^{-4}
7	GO:BP	GO:0006353	DNA-templated transcription termination	9.098×10^{-4}
8	GO:BP	GO:0071034	CUT catabolic process	9.923×10^{-3}
9	GO:BP	GO:0071042	nuclear polyadenylation-dependent mRNA catabol...	1.753×10^{-2}
10	GO:BP	GO:0000184	nuclear-transcribed mRNA catabolic process, nons...	3.238×10^{-2}
11	GO:CC	GO:0005730	nucleolus	3.360×10^{-48}

Notes – Term IDs listed in overview are the most significant contributors to the enriched term

- Overall trends of graphs are similar

MF

- RNA binding, catalytic activity, acting on RNA, and N-Methyltransferase activity significantly affected in both

BP

CC

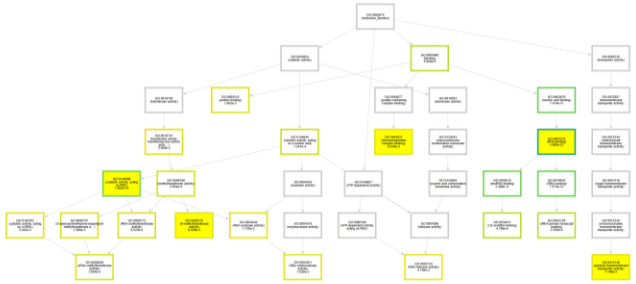
- Nucleolus activity is significantly affected by cold shock in the profile of both strains
- Organellar ribosome & endonuclease complex only affected in WT strain

Profile 45

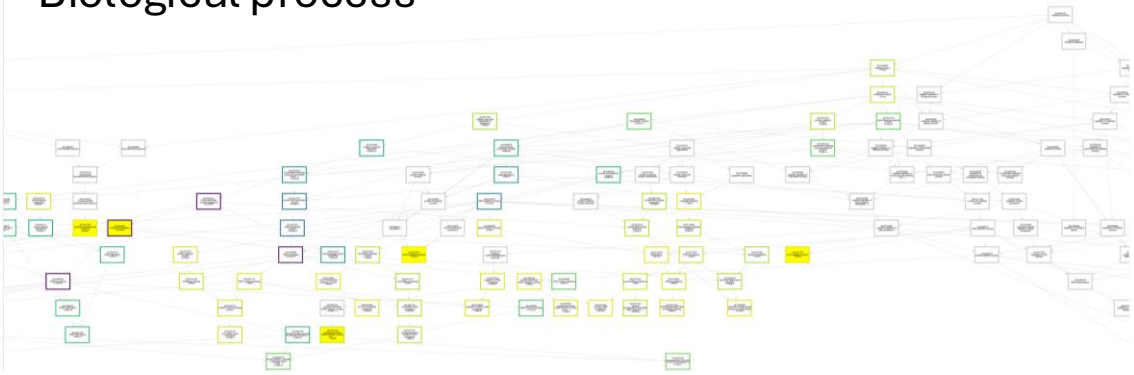
Biological process is the largest umbrella of cellular processes in both strains

WT

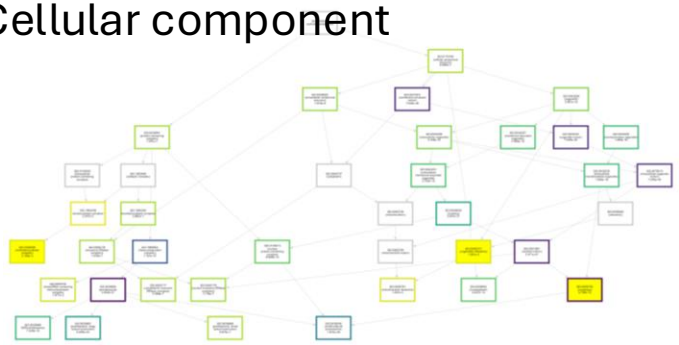
Molecular function



Biological process



Cellular component

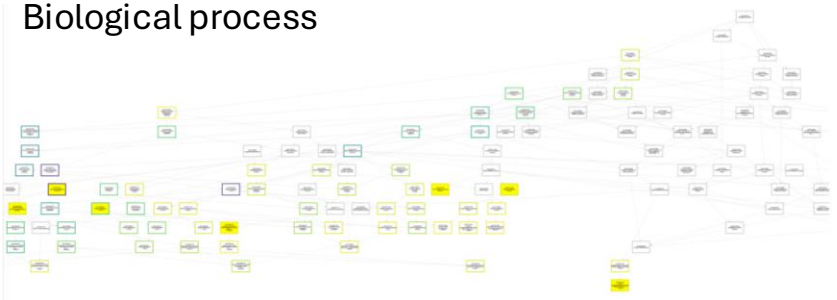


- dCIN5

Molecular function



Biological process



Cellular component



Profile 45 – general conclusions

- WT is more affected
  - When Cin5 is deleted there are functions that may be not present therefore not affected by cold shock?

Significant gene functions that are conserved across strains of *S. cerevisiae*